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Single line drive circuit enabling identical actuator orientation

Abstract

A driving circuit is described. The driving circuit includes: a first piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connected to a first ground at a first terminal and the second electrode configured to connected to an amplifier at a second terminal. The driving circuit includes a second piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connected to a control signal at a first terminal and the second electrode connected to a second ground at a second terminal. And, the first terminal of the first piezoelectric actuator and the first terminal of the second piezoelectric actuator are configured such that the first piezoelectric actuator and the second piezoelectric actuator are symmetrical and have similar polarity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/116,722 filed on Nov. 20, 2020, which is hereby incorporated by reference in its entirety.

FIELD

(1) This disclosure relates to the field of piezoelectric actuators. More particularly, this disclosure relates to improving performance of piezoelectric actuators.

DESCRIPTION OF RELATED ART

(2) A hard disk drive is known as a type of information storage device. A hard disk drive typically includes one or more magnetic disks rotatably mounted in association with a spindle and one or more actuator assemblies for positioning a magnetic transducer, or head, relative to concentric circular data tracks on a magnetic medium-bearing surface of each disk.

(3) The recording densities of hard disk drives have been increasing with advances in personal computers, so that data tracks are becoming increasingly more densely positioned on the disks, and the tracks themselves are becoming physically narrower. As a result, maintaining the transducer or head in an accurate track-following position for purposes of reading and writing, is becoming more difficult. To accommodate the needed increasingly finer adjustments in the position of the magnetic head, a precision positioning actuator has been introduced. For such an actuator, a single piezoelectric actuator assembly is generally employed. The piezoelectric actuator forms a part of a head gimbal assembly which moves the head transverse to a track to provide fine radial positioning of a head in reference to a track.

SUMMARY

(4) A driving circuit is described. The driving circuit includes: a first piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connected to a first ground at a first terminal and the second electrode configured to connected to an amplifier at a second terminal. The driving circuit includes a second piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connected to a control signal at a first terminal and the second electrode connected to a second ground at a second terminal. And, the first terminal of the first piezoelectric actuator and the first terminal of the second piezoelectric actuator are configured such that the first piezoelectric actuator and the second piezoelectric actuator are symmetrical and have similar polarity.

(5) In some examples of the driving circuit, the first piezoelectric actuator includes a first piezoelectric element, a second piezoelectric element, and a third piezoelectric element. The first piezoelectric element may include a top surface partially covered by a portion of the first electrode and a portion of the second electrode, where the second electrode covers more of the top surface of the first piezoelectric element than the first electrode. The first piezoelectric element and the second piezoelectric element are partially separated by the first electrode, which is connected to the first ground. The second piezoelectric element and the third piezoelectric element may be partially separated by the second electrode. The third piezoelectric element may include a bottom surface partially covered by a portion of the first electrode and a portion of the second electrode, where the second electrode covers less of the bottom surface of the third piezoelectric element than the first electrode.

(6) In some examples of the driving circuit, the second piezoelectric actuator includes a first piezoelectric element, a second piezoelectric element, and a third piezoelectric element. The first piezoelectric element may include a top surface partially covered by a portion of the first electrode and a portion of the second electrode, where the second electrode covers more of the top surface of the first piezoelectric element than the first electrode. The first piezoelectric element and the second piezoelectric element are partially separated by the first electrode. The second piezoelectric element and the third piezoelectric element may be partially separated by second electrode. The third piezoelectric element may have a bottom surface partially covered by a portion of the first electrode and a portion of the second electrode, where the second electrode covers less of the bottom surface of the third piezoelectric element than the first electrode.

(7) In some examples of the driving circuit, the first piezoelectric actuator is configured to be driven in response to a control signal applied via the second terminal from the amplifier. In some examples of the driving circuit, the second piezoelectric actuator is configured to be driven in response to a control signal applied via the first terminal from the amplifier. In some examples, the

driving circuit also includes a head slider. The similar polarity and the symmetry of the first piezoelectric actuator and the second piezoelectric actuator enable fine radial positioning of the head slider.

(8) A suspension device is also provided. The suspension device includes a driving circuit, which includes a first piezoelectric actuator comprising at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode connected to a first ground at a first terminal and the second electrode connected to an amplifier at a second terminal. The driving circuit also includes a second piezoelectric actuator comprising at least one piezoelectric element disposed between a first electrode and a second electrode. The first electrode is connected to the amplifier at a first terminal and the second electrode connected to a second ground at a second terminal. The first terminal of the first piezoelectric actuator and the first terminal of the second piezoelectric actuator are positioned such that the first piezoelectric actuator and the second piezoelectric actuator have similar polarity and orientation.

(9) While multiple examples are disclosed, still other examples of the present disclosure will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative examples of this disclosure. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the above-recited and other advantages and features of the disclosure can be obtained, a more particular description of the principles described above will be rendered by reference to specific examples illustrated in the appended drawings. These drawings depict only example aspects of the disclosure and are therefore not to be considered as limiting of its scope. The principles are described and explained with additional specificity and detail using the following drawings.

(2) FIG. 1 illustrates a side sectional view of a first piezoelectric actuator for a driving circuit, in accordance with an example of the disclosure.

(3) FIG. 2 illustrates a side sectional view of a second piezoelectric actuator, in accordance with an example of the disclosure.

(4) FIG. 3 illustrates a side sectional view of a first piezoelectric actuator, in accordance with an example of the disclosure.

(5) FIG. 4 illustrates a side sectional view of a second piezoelectric actuator, in accordance with an example of the disclosure.

(6) FIG. 5 illustrates a driving circuit architecture, in accordance with an example of the disclosure.

(7) FIG. 6 illustrates a side sectional view of a first piezoelectric actuator of the driving circuit architecture of FIG. 5, in accordance with an example of the disclosure.

(8) FIG. 7 illustrates a side sectional view of a second piezoelectric actuator of the driving circuit architecture of FIG. 5, in accordance with an example of the disclosure.

(9) FIG. 8 is a graph of the PZT frequency response function of a suspension incorporating the driving circuit architecture, in accordance with an example of the disclosure.

(10) FIG. 9 is a graph of the PZT frequency response function of a suspension incorporating the driving circuit architecture, in accordance with an example of the disclosure.

(11) FIG. 10 is a table illustrating a stroke of a suspension incorporating the driving circuit according to some embodiments described herein, in accordance with an example of the disclosure.

(12) FIG. 11 is a graph comparing the PZT frequency response function of a suspension incorporating, according to a simulation.

DETAILED DESCRIPTION

(13) FIG. 1 illustrate a side sectional view of a first piezoelectric actuator **120** having a first polarity. The first piezoelectric actuator **120** has a first electrode **112** that is configured to connected to a first reference potential, for example, a zero potential (e.g., a ground). The first piezoelectric actuator **120** also has a second electrode **114** is configured to connect to a control signal **130**. The first piezoelectric actuator **120** includes a first piezoelectric element **122**, a second piezoelectric element **124**, and a third piezoelectric element **126**. The first piezoelectric element **122** has a top surface partially covered by a portion of the first electrode **112** and a portion of the second electrode **114**. The second electrode **114** covers more of the top surface of the first piezoelectric element **122** than the first electrode **112**. The first piezoelectric element **122** and the second piezoelectric element **124** are partially separated by the first electrode **112**. Moreover, the second piezoelectric element **124** and the third piezoelectric element **126** are partially separated by second electrode **114**. The third piezoelectric element **126** has a bottom surface partially covered by a portion of the first electrode **112** and a portion of the second electrode **114**. The second electrode **114** covers less of the bottom surface of the third piezoelectric element **126** than the first electrode **112**.

(14) Under the illustrated poling directions in FIG. 1, when a positive voltage is applied and the electric field is in the same direction as the poling direction across the first electrode **112** and the second electrode **114**, all three piezoelectric elements **122**, **124**, and **126** expand in the longitudinal direction.

(15) FIG. 2 illustrates a side sectional view of a second piezoelectric actuator **125** having a second polarity opposite the first polarity. The second piezoelectric actuator **125** has a first electrode **116** that is configured to connect to a first reference potential, for example, a zero potential (e.g., the second ground **180**). The second piezoelectric actuator **125** also has a second electrode **118** is connected to a control signal **130**. The second piezoelectric actuator **125** includes a first piezoelectric element **132**, a second piezoelectric element **134**, and a third piezoelectric element **136**. The first piezoelectric element **132** has a top surface partially covered by a portion of the first electrode **116** and a portion of the second electrode **118**.

(16) The second electrode **118** covers more of the top surface of the first piezoelectric element **132** than the first electrode **116**. The first piezoelectric element **132** and the second piezoelectric element **134** are partially separated by the first electrode **116**. Moreover, the second piezoelectric element **134** and the third piezoelectric element **136** are partially separated by second electrode **118**. The third piezoelectric element **136** has a bottom surface partially covered by a portion of the first electrode **116** and a portion of the second electrode **118**. The second electrode **118** covers less of the bottom surface of the third piezoelectric element **136** than the first electrode **116**.

(17) Under the illustrated poling directions as illustrated in FIG. 2, when a positive voltage is applied and the electric field is in the opposite direction of the poling direction across the first electrode **116** and the second electrode **118**, all three piezoelectric elements **132**, **134**, and **136** contract in the longitudinal direction.

(18) FIG. 3 illustrates a side sectional view of the first piezoelectric actuator **220** for a driving circuit. The first piezoelectric actuator **220** has a first electrode **212** that is configured to connect to a first reference potential, for example, a zero potential (e.g., a ground). The first piezoelectric actuator **220** also has a second electrode **214** connected to a control signal **230**. The first piezoelectric actuator **220** includes a first piezoelectric element **222**, a second piezoelectric element **224**, and a third piezoelectric element **226**. The first piezoelectric element **222** has a top surface partially covered by a portion of the first electrode **212** and a portion of the second electrode **214**.

(19) The second electrode **214** covers more of the top surface of the first piezoelectric element **222** than the first electrode **212**. The first piezoelectric element **222** and the second piezoelectric element **224** are partially separated by the first electrode **212**. Moreover, the second piezoelectric element **224** and the third piezoelectric element **226** are partially separated by second electrode **214**. The third piezoelectric element **226** has a bottom surface partially covered by a portion of the

first electrode **212** and a portion of the second electrode **214**. The second electrode **214** covers less of the bottom surface of the third piezoelectric element **226** than the first electrode **212**.

(20) Under the illustrated poling directions in FIG. 3, when a positive voltage is applied and the electric field is in the same direction as the poling direction across the first electrode **212** and the second electrode **214**, all three piezoelectric elements **222**, **224**, and **226** expand in the longitudinal direction.

(21) FIG. 4 illustrates a side sectional view of the second piezoelectric actuator **225** of the driving circuit, in accordance with an example of the disclosure. The second piezoelectric actuator **225** has a first electrode **216** that is connected to a first reference potential, for example, a zero potential (e.g., a ground). The second piezoelectric actuator **225** also has a second electrode **218** connected to the control signal **230**. The second piezoelectric actuator **225** includes a first piezoelectric element **322**, a second piezoelectric element **324**, and a third piezoelectric element **326**. The first piezoelectric element **322** has a top surface partially covered by a portion of the first electrode **216** and a portion of the second electrode **218**.

(22) In contrast to the first piezoelectric actuator **220** of FIG. 3, the second electrode **218** covers less of the top surface of the first piezoelectric element **322** than the first electrode **216**. This is because the second piezoelectric actuator **225** is rotated 180 degrees from the orientation of the second piezoelectric actuator **125** of FIG. 2. The first piezoelectric element **322** and the second piezoelectric element **324** are partially separated by the second electrode **218**. Moreover, the second piezoelectric element **324** and the third piezoelectric element **326** are partially separated by first electrode **216**. The third piezoelectric element **326** has a bottom surface partially covered by a portion of the first electrode **216** and a portion of the second electrode **218**. The second electrode **218** covers more of the bottom surface of the third piezoelectric element **326** than the first electrode **216**.

(23) Under the illustrated poling directions in FIG. 4, when a positive voltage is applied such that the electric field is in an opposite direction of the poling direction across the first electrode **216** and the second electrode **218**, all three piezoelectric elements **322**, **324**, and **326** contract in the longitudinal direction.

(24) FIG. 5 is a driving circuit **500**, in accordance with an example of the disclosure. The driving circuit **500** includes a first piezoelectric actuator **520** and a second piezoelectric actuator **525** configured to move a head slider **540** of a suspension for a hard disk drive. The first piezoelectric actuator **520** and the second piezoelectric actuator **525** are each electrically coupled with the driving circuit **500**. As a result, the first piezoelectric actuator **520** and the second piezoelectric actuator **525** are responsive to an applied voltage control signal applied to affect a desired physical displacement of a head slider using techniques including those known in the prior art. This is discussed in greater detail below with respect to FIGS. 6-7.

(25) The driving circuit **500** is configured to be incorporated in a suspension device of a hard disk drive, such as a head gimbal assembly. The first piezoelectric actuator **520** includes a first terminal **521** (or reference terminal), and a second terminal **523**. The first piezoelectric actuator **520** is connected to a first ground contact **570** at the first terminal **521**. The first piezoelectric actuator **520** is also connected to a control signal **530** at the second terminal **523**. The second piezoelectric actuator **525** includes a first terminal **527** (e.g., a reference terminal), and a second terminal **529**. The second piezoelectric actuator **525** is connected to a second ground contact **580** at the second terminal **529**. The second piezoelectric actuator **525** is also connected to the control signal **530** at the first terminal **527**. In other words, the first and second terminals **527**, **529** of the second piezoelectric actuator **525** are switched and the first and second piezoelectric actuators **520**, **525** have the same polarity and orientation. And, the active portions **528**, **538** are arranged such that they are symmetrical. This is further illustrated in FIGS. 6-7.

(26) FIG. 6 illustrates a side sectional view of the first piezoelectric actuator **520** of the driving circuit architecture **500**, in accordance with an example of the disclosure. The first piezoelectric

actuator **520** has a first electrode **512** that is connected to a first reference potential, for example, zero potential (e.g., through the first ground contact **570**). The first piezoelectric actuator **520** also has a second electrode **514** connected to the control signal **530**. The first piezoelectric actuator **520** includes a first piezoelectric element **522**, a second piezoelectric element **524**, and a third piezoelectric element **526**. The first piezoelectric element **522** has a top surface partially covered by a portion of the first electrode **512** and a portion of the second electrode **514**.

(27) The second electrode **514** covers more of the top surface of the first piezoelectric element **522** than the first electrode **512**. The first piezoelectric element **522** and the second piezoelectric element **524** are partially separated by the first electrode **512**, which is grounded. Moreover, the second piezoelectric element **524** and the third piezoelectric element **526** are partially separated by second electrode **514**. The third piezoelectric element **526** has a bottom surface partially covered by a portion of the first electrode **512** and a portion of the second electrode **514**. The second electrode **514** covers less of the bottom surface of the third piezoelectric element **526** than the first electrode **512**, which is grounded. The first piezoelectric actuator **520** includes an active portion **528** that includes a portion of the first piezoelectric element **522** between the second electrode **514** and the first electrode **512**; a portion of the second piezoelectric element **524** between the first electrode **512** and the second electrode **514**; and a portion of the third piezoelectric element **526** between the second element **514** and the first element **512**.

(28) Under the illustrated poling directions in FIG. **6**, when a positive voltage is applied such that the electric field is in the same direction as the poling direction across the first electrode **512** and the second electrode **514**, all three piezoelectric elements **522**, **524**, and **526** expand in the longitudinal direction.

(29) FIG. **7** illustrates a side sectional view of the second piezoelectric actuator **525** of the driving circuit **500**, in accordance with an example of the disclosure. The second piezoelectric actuator **525** has a first electrode **516** that is connected to the control signal **530**. The second piezoelectric actuator **525** also has a second electrode **518** connected to a first reference potential, for example, zero potential (e.g., through the second ground contact **580**). The second piezoelectric actuator **525** includes a first piezoelectric element **532**, a second piezoelectric element **534**, and a third piezoelectric element **536**. The first piezoelectric element **532** has a top surface partially covered by a portion of the first electrode **516** and a portion of the second electrode **518**.

(30) The second (grounded) electrode **518** covers more of the top surface of the first piezoelectric element **532** than the first electrode **516**. The first piezoelectric element **532** and the second piezoelectric element **534** are partially separated by the first electrode **516**. Moreover, the second piezoelectric element **534** and the third piezoelectric element **536** are partially separated by second electrode **518**. The third piezoelectric element **536** has a bottom surface partially covered by a portion of the first electrode **516** and a portion of the second electrode **518**. The second electrode **518** covers less of the bottom surface of the third piezoelectric element **536** than the first electrode **516**. The second piezoelectric actuator **525** includes an active portion **538** that includes a portion of the piezoelectric element **532** between the second electrode **518** and the first electrode **516**; a portion of the second piezoelectric element **534** between the first electrode **516** and the second electrode **518**; and a portion of the third piezoelectric element **536** between the second element **518** and the first element **516**.

(31) Under the illustrated poling directions in FIG. **7**, when a positive voltage is applied such that the electric field is in opposite direction of poling direction across the first electrode **516** and the second electrode **518**, all three piezoelectric elements **532**, **534**, and **536** contract in the longitudinal direction.

(32) Referring back to FIG. **5**, the first piezoelectric actuator **520** and the second piezoelectric actuator **525** are similarly poled. Moreover, the structures of the first piezoelectric actuator **520** and the second piezoelectric actuator **525** are arranged in a symmetrical orientation and each having the same polarity. The first piezoelectric actuator **520** is configured to be driven in response to a control

signal **530** applied via the second (input) terminal **523**. Similarly, the second piezoelectric actuator **525** is configured to be driven in response to a control signal **530** applied via the first (input) terminal **527**. The control signals affect mechanical deformation of the piezoelectric elements, as discussed above, of the first and second piezoelectric actuators **520**, **525**. The similar polarity and the symmetry of the first piezoelectric actuator **520** and the second piezoelectric actuator **525** enable fine radial positioning of the head slider **540** in direction **510**. The similar polarity and the symmetry of the first piezoelectric actuator **520** and the second piezoelectric actuator **525** also improve sway mode gain and torsion mode gain sensitivity, as compared to current systems that rely on actuators having an opposite orientation and asymmetric active portions.

(33) FIG. **8** is a graph **600** of the PZT frequency response function (0-80 kHz) of a suspension incorporating driving circuit according to some embodiments described herein, according to a simulation. FIG. **9** is a graph **700** of the PZT frequency response function (0-30 kHz) of a suspension incorporating driving circuit according to some embodiments described herein, according to a simulation. As indicated in FIG. **9**, the suspension exhibited torsion mode gain sensitivity of 1 dB and sway mode gain sensitivity of 3 dB. The lower sway mode gain and torsion mode gain sensitivity increase head positioning control loop bandwidth, which translates to both lower data seek times and lower susceptibility to vibrations.

(34) FIG. **10** is a table **800** illustrating the stroke of a suspension incorporating the driving circuit according to some embodiments described herein compared to a suspension incorporating current techniques including asymmetric arrangement of the active portions and opposite actuator orientation, according to a simulation. As illustrated herein, the suspension incorporating the driving circuit according to some embodiments described herein with actuators with the same orientation and symmetric active portions exhibited a stroke of 10.05 nm/V. In comparison, the suspension incorporating the asymmetric arrangement of the active portions of the actuators exhibited a stroke of 8.99 nm/V. Therefore, the suspension incorporating the driving circuit according to some embodiments described herein exhibited a stroke increase. For the example illustrated in FIG. **10**, the driving circuit according to embodiments described herein exhibited stroke increase of 11% compared to the suspension incorporating the asymmetric arrangement and opposite actuator orientation.

(35) FIG. **11** is a graph **900** comparing the PZT frequency response function of a suspension incorporating driving circuit **500** to the PZT frequency response function of a suspension incorporating current techniques including asymmetric arrangement of the active portions and opposite actuator orientation, according to a simulation. As illustrated in FIG. **11**, the suspension incorporating driving circuit according to some embodiments described herein exhibited lower sway mode gain and torsion mode gain sensitivity.

(36) It will be understood that the terms “generally,” “approximately,” “about,” “substantially,” and “coplanar” as used within the specification and the claims herein allow for a certain amount of variation from any exact dimensions, measurements, and arrangements, and that those terms should be understood within the context of the description and operation of the present disclosure.

(37) It will further be understood that terms such as “top,” “bottom,” “above,” and “below” as used within the specification and the claims herein are terms of convenience that denote the spatial relationships of parts relative to each other rather than to any specific spatial or gravitational orientation. Thus, the terms are intended to encompass an assembly of component parts regardless of whether the assembly is oriented in the particular orientation shown in the drawings and described in the specification, upside down from that orientation, or any other rotational variation.

(38) All features disclosed in the specification, including the claims, abstract, and drawings, and all the steps in any method or process disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. Each feature disclosed in the specification, including the claims, abstract, and drawings, can be replaced by alternative features serving the same, equivalent, or similar purpose, unless expressly stated

otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

(39) It will be appreciated that the term “example” as used herein should not be construed to mean that only a single example having a single essential element or group of elements is presented. Similarly, it will also be appreciated that the term “present disclosure” encompasses a number of separate innovations which can each be considered separate examples. Although the present disclosure has thus been described in detail with regard to the preferred examples and drawings thereof, it should be apparent to those skilled in the art that various adaptations and modifications of the present disclosure may be accomplished without departing from the spirit and the scope of the present disclosure. Accordingly, it is to be understood that the detailed description and the accompanying drawings as set forth hereinabove are not intended to limit the breadth of the present disclosure, which should be inferred only from the following claims and their appropriately construed legal equivalents.

Claims

1. A driving circuit comprising: a first piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connected to a first ground at a first terminal and the second electrode configured to connected to an amplifier at a second terminal and wherein the first piezoelectric actuator includes a first piezoelectric element, the first piezoelectric element having a top surface partially covered by a portion of the first electrode and a portion of the second electrode, and the second electrode covers more of the top surface of the first piezoelectric element than the first electrode; and a second piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connected to a control signal at a first terminal and the second electrode connected to a second ground at a second terminal, and wherein the second piezoelectric actuator includes a first piezoelectric element, the first piezoelectric element having a top surface partially covered by a portion of the first electrode and a portion of the second electrode, and the second electrode covers more of the top surface of the first piezoelectric element than the first electrode, the first terminal of the first piezoelectric actuator and the first terminal of the second piezoelectric actuator are configured such that the first piezoelectric actuator and the second piezoelectric actuator are symmetrical and have similar polarity.
2. The driving circuit of claim 1, wherein the first piezoelectric actuator further includes a second piezoelectric element and a third piezoelectric element.
3. The driving circuit of claim 2, wherein the first piezoelectric element and the second piezoelectric element are partially separated by the first electrode, which is connected to the first ground.
4. The driving circuit of claim 2, wherein the second piezoelectric element and the third piezoelectric element are partially separated by the second electrode.
5. The driving circuit of claim 2, wherein the third piezoelectric element includes a bottom surface partially covered by a portion of the first electrode and a portion of the second electrode, the second electrode covers less of the bottom surface of the third piezoelectric element than the first electrode.
6. The driving circuit of claim 1, wherein the second piezoelectric actuator further includes a second piezoelectric element and a third piezoelectric element.
7. The driving circuit of claim 6, wherein the first piezoelectric element and the second piezoelectric element are partially separated by the first electrode.
8. The driving circuit of claim 6, wherein the second piezoelectric element and the third piezoelectric element are partially separated by second electrode.
9. The driving circuit of claim 6, wherein the third piezoelectric element has a bottom surface

partially covered by a portion of the first electrode and a portion of the second electrode, the second electrode covers less of the bottom surface of the third piezoelectric element than the first electrode.

10. The driving circuit of claim 1, wherein the first piezoelectric actuator is configured to be driven in response to the control signal applied via the second terminal.

11. The driving circuit of claim 1, wherein the second piezoelectric actuator is configured to be driven in response to the control signal applied via the first terminal.

12. The driving circuit of claim 1, comprising a head slider, wherein the first piezoelectric actuator and the second piezoelectric actuator that are symmetrical and have similar polarity enable fine radial positioning of the head slider.

13. A suspension device comprising: a driving circuit comprising: a first piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connect to a first ground at a first terminal and the second electrode configured to connect to a control signal at a second terminal and wherein the first piezoelectric actuator includes a first piezoelectric element, the first piezoelectric element having a top surface partially covered by a portion of the first electrode and a portion of the second electrode, and the second electrode covers more of the top surface of the first piezoelectric element than the first electrode; and a second piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connect to a control signal at a first terminal and the second electrode configured to connect to a second ground at a second terminal, and wherein the second piezoelectric actuator includes a first piezoelectric element, the first piezoelectric element having a top surface partially covered by a portion of the first electrode and a portion of the second electrode, and the second electrode covers more of the top surface of the first piezoelectric element than the first electrode, the first terminal of the first piezoelectric actuator and the first terminal of the second piezoelectric actuator are configured such that the first piezoelectric actuator and the second piezoelectric actuator are symmetrical and have similar polarity.

14. The suspension device of claim 13, wherein the first piezoelectric actuator further includes a second piezoelectric element and a third piezoelectric element.

15. The suspension device of claim 13, wherein the second piezoelectric actuator further includes a second piezoelectric element and a third piezoelectric element.

16. The suspension device of claim 13, wherein the first piezoelectric actuator is configured to be driven in response to the control signal applied via the second terminal.

17. The suspension device of claim 13, wherein the second piezoelectric actuator is configured to be driven in response to the control signal applied via the first terminal.

18. The suspension device of claim 13, comprising a head slider, wherein the first piezoelectric actuator and the second piezoelectric actuator that are symmetrical and have similar polarity enable fine radial positioning of the head slider.

19. A suspension device comprising: a driving circuit comprising: a first piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connect to a first ground at a first terminal and the second electrode configured to connect to a control signal at a second terminal and wherein the first piezoelectric actuator includes a first piezoelectric element, the first piezoelectric element having a top surface partially covered by a portion of the first electrode and a portion of the second electrode, and the first electrode covers more of the top surface of the first piezoelectric element than the second electrode; and a second piezoelectric actuator including at least one piezoelectric element disposed between a first electrode and a second electrode, the first electrode configured to connect to a control signal at a first terminal and the second electrode configured to connect to a second ground at a second terminal, and wherein the second piezoelectric actuator includes a first piezoelectric element, the first piezoelectric element having a top surface partially covered by a

portion of the first electrode and a portion of the second electrode, and the first electrode covers more of the top surface of the first piezoelectric element than the second electrode, the first terminal of the first piezoelectric actuator and the first terminal of the second piezoelectric actuator are configured such that the first piezoelectric actuator and the second piezoelectric actuator are symmetrical and have similar polarity.

20. The suspension device of claim 19, wherein the second piezoelectric actuator further includes a second piezoelectric element and a third piezoelectric element.

21. The suspension device of claim 20, wherein the third piezoelectric element has a bottom surface partially covered by a portion of the first electrode and a portion of the second electrode, and the second electrode covers more of the bottom surface of the third piezoelectric element than the first electrode.
